

Steroid Induced Ocular Hypertension and Treatment in A 40 Year General Medical Practitioner.

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ABSTRACT

Steroids are commonly used in clinical practice to combat allergy, vascular diseases, and other inflammatory processes. They are administered through systemic or local routes and get to the eyes to elicit the response of raised intraocular pressure (IOP), and in chronic states, lead to glaucoma in susceptible individuals. A 40-year-old medical practitioner with severe bilateral conjunctivitis was treated with anti-allergic and nonsteroidal anti-inflammatory drugs (NSAID) elsewhere without remission. On review, we added a steroidal eye drop; dexamethasone and other drugs to the medications she was using. Five weeks after, the patient reported significant improvement in condition, but IOP rose from 17mmHg in both eyes to 39, 37mmHg right (RE) and left (LE) eyes, respectively. IOP lowering drugs were added, and subsequently, steroids were discontinued by tapering. The IOP returned to baseline in 7 weeks, and IOP lowering drugs were stopped at 9 weeks. Ophthalmologists must look out for this group of patients on long term steroid use to avoid the development of glaucoma.

Keywords: Antiglaucoma, Conjunctivitis, Glaucoma, Ocular hypertension, Steroid, Visual loss.

INTRODUCTION

Steroids are used in clinical practice to treat a wide range of disease conditions and in ophthalmic practice in treating both extraocular and especially intraocular inflammatory processes because of the need to prevent undesirable effects of inflammation in the eye.^{1,2} Duke elder et al³ were the first in a series of cases to document its use in treating various ocular disease conditions. However, they were discovered to have ocular side effects such as cataract and steroid induced ocular hypertension (SIOH) and eventual glaucoma that could compromise vision.⁴ The case presented is steroid induced ocular hypertension seen in a patient treated with steroids for extraocular inflammation.

The mechanism of steroid induce pressure is not fully understood, but there are theories that explains it. In the gene theory, Rozsa et al⁵ reported a more than a 100-fold increase in expression of the myocilin MYOC

gene on exposure to steroids. The increase receptor expression of this response genes MYOC or Trabecular meshwork-induced glucocorticoid response (TIGR)] gene at locus GLC1A, provides the signaling which leads primarily to reduce facility aqueous outflow with raised IOP.⁶ This is further explained by theories that propose that there is an accumulation of polymerized glycosaminoglycans (GAGs), which become hydrated and swollen, leading to an increased outflow resistance in the trabecular meshwork.^{7,11} Another, proposes an increased expression of extracellular matrix protein and inhibition of the phagocytotic properties of endothelial cells lining the trabecular meshwork, which leads to accumulation of aqueous debris, increasing resistance.⁷⁻⁹ Again, steroids decrease the synthesis of prostaglandin, which regulates the aqueous outflow.¹⁰

THE CASE

A 40-year-old general medical practitioner who had no family history of glaucoma presented with itching, pains, photophobia, tearing, redness, and grittiness in

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both eyes of 3 weeks duration. She had earlier been placed on drops Ketotifen 0.025% tds and diclofenac 0.1% for 2 weeks by an optometrist without improvement. Key findings were: Unaided visual acuity (VA) 6/9, which improved to 6/5 with Plano-50×90; IOP was 17mmHg in both eyes (BE); conjunctival injection with mild chemosis; giant papillae; trantas spots; and CDR 0.4 and 0.35 R and LE, respectively. After assessing, a diagnosis of severe acute allergic conjunctivitis was made. She was placed on a Steroid/antibiotic combination of Maxitrol bd, Zinchlor (Boric acid, Zinc sulphate, Naphazoline hydrochloride, Chlorpheniramine maleate and Sodium chloride) Qds, and tabs loratadine I daily for 7 days. On review at 2 weeks, the condition had worsened with increased redness, lacrimation, and grittiness, but the measured IOP was unchanged. The steroid frequency was increased to qds, added drops Opticrom (Sodium cromoglycate) Qds, Diclofenac bd, and tabs of Cataflam 50mg bd for 5 days.

At 4 weeks, patient's symptoms had improved remarkably, and because there was no change in IOP, she was maintained on the same medications but reduced the steroid to tds for a week and then bd for another to review in 2 weeks. At 5 weeks of steroid use, there was further improvement in the patient's condition, but IOP had increased to 39 and 37mmHg in the right and left eyes.

Treatment was instituted by introducing IOP lowering drugs: tabs Acetazolamide 250 mg bd for 5 days and drops Cusopt (Dorzolamide hydrochloride and Timolol maleate) bd till next review at 7 weeks, when IOP had returned to 12mmHg in BE. The patient still complained of mild itching but was maintained on Zinchlor and tabs Lorantidine while the pressure-lowering drugs were stopped after another 2 weeks.

DISCUSSION

Conjunctival allergic conditions are managed with antihistamines, mast cell

stabilizers, and in rare cases with steroids.² Beyond allergy, steroids are commonly used in ophthalmic practice to treat other intra and extraocular inflammatory conditions by modulating the immune responses. In case of the patient under review, topical steroid treatment was indicated because of severity of disease and non-response to anti-allergic medications.

In the reviewed case, IOP rose after administration of topical steroids and the raised pressure is known to be a major risk factor for glaucoma, a type of optic neuropathy that causes irreversible visual loss and blindness.¹¹ Similarly, Wood⁴ reported and was the first to report raise IOP with steroid use but in the year 1950 Becker and mills¹² showed that the effect was more among glaucoma patients, followed by glaucoma suspects and then least in non-glaucoma patients. Becker and mills' findings are consistent with reports that glaucoma and family history are risk factor for SIOH, but these factors were not found in the patient.

Other factors identified in the patient are the steroid response status and drug dependency factors. The individual steroid response status that raised the IOP to 39 and 37mmHg suggest the patient is a high steroid responder, which is inherent genetically. This category of people constitutes 4-6% of the general population and has an IOP greater than 31mmHg or a rise of more than 15mmHg from baseline. Other categories are the moderate responders, who are about 1/3 of the population and are found to develop an IOP between 25-31 mm Hg or a rise of 6-15 mm Hg from baseline. The last group is the non-responders, who are about 2/3 of the population, with an IOP less than 20 mm Hg or a rise of less than 6 mm Hg from baseline.

The drug dependent factors that predisposed the patient are: potency, dexamethasone is a potent steroid; dosage, used in high dose QDS; and duration, for 5 weeks. All 3 factors; potency, dosage and duration of use are known to affect response.¹²

Furthermore, this patient is classified an early responder base on duration which normally occurs at 3-6 weeks of continuous use^{13,14} and is associated with topical route of administration, as found with the patient.¹⁵ In contrast, delayed response is mainly caused by the systemic route of administration and takes several months. Other factors, though not found in the patient, are: age, mostly young children and older adults,^{13,16} high myopia, type 1 diabetes mellitus, and rheumatoid arthritis.¹⁶

In the treatment of the patient, IOP lowering medications were prescribed immediately and is one of the modalities.¹⁷ In the study by Pradhya,¹⁷ it reported how he managed steroid-induced pressures: withdrawal of steroids, with antiglaucoma medications, surgery alone, and use of surgery with drainage devices. It achieved IOP control by the withdrawal of steroids (9 eyes), with antiglaucoma medications (27 eyes), trabeculectomy (57 eyes), and glaucoma drainage device (1 eye).¹⁷ The study was carried out on 1423 patients with VKC who were on steroids, among whom 92 eyes of 47 patients had steroid-induced glaucoma, with a prevalence 3.3%. Their report indicates that the first step in treatment involves withdrawal of steroid. This alone, is reported to normalize IOP levels within 10 days to 3 weeks after discontinuation of therapy.¹⁸ The patient in review, the steroid treatment could not be discontinued because it is the main indication in severe allergy and the patient did not do well with antiallergic alone till it was introduced. Again, sudden and complete withdrawal of steroid may not be possible in some disease conditions as obtains in uveitis, and in the present case, to prevent a rebound often experienced when there is a sudden cessation. Instead, pressure reducing drugs were commenced, then the steroid was maintained and tapered later, which was still able to achieve a smooth improvement in condition with a stable IOP control.

An alternative to complete withdrawal of dexamethasone was to switch to a less potent steroid preparation. Fluorometholone 0.1%, has less effect on IOP.¹⁹ This would have taken a longer time than the 4 week we achieved control with antiglaucoma drugs. Additionally, even after withdrawal of steroid and additional medical or surgical therapy to control the IOP, up to 3% of steroid responders may have irreversible elevations of IOP.²⁰

CONCLUSION

Ophthalmologists must be cautious in the use of steroids and must be on the lookout for responders to prevent the development of steroid-induced glaucoma.

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